```
0    BEGIN PGM 1 MM
1    BLK FORM 0.1 Z    X-25    Y-25    Z-30
2    BLK FORM 0.2    X+25    Y+25    Z+0
3    CYCL DEF 247 DATUM SETTING ~
       Q339=+1        ;DATUM NUMBER
4    TOOL CALL 5 Z S1000
5    LBL 2
6    CYCL DEF 7.0 DATUM SHIFT
7    CYCL DEF 7.1    X+0
8    CYCL DEF 7.2    Y+0
9    CYCL DEF 7.3    Z+0
10 CYCL DEF 19.0 WORKING PLANE
11 CYCL DEF 19.1    A+0    B+0    C+0
12 L    B+Q121    C+Q122 FMAX
13 LBL 0
14 L    Z+200 R0 F2000 M3
15 LBL 3
16 CYCL DEF 7.0 DATUM SHIFT
17 CYCL DEF 7.1 IX+0
18 CYCL DEF 7.2 IY-25
19 CYCL DEF 7.3 IZ+0
20 CYCL DEF 19.0 WORKING PLANE
21 CYCL DEF 19.1    A-5    B+0    C+0
22 CYCL DEF 7.0 DATUM SHIFT
23 CYCL DEF 7.1 IX+0
24 CYCL DEF 7.2 IY+25
25 CYCL DEF 7.3 IZ+0
26 L    B+Q121    C+Q122 FMAX
27 LBL 0
28 CYCL DEF 251 RECTANGULAR POCKET ~
       Q215=+0        ;MACHINING OPERATION ~
       Q218=+52       ;FIRST SIDE LENGTH ~
       Q219=+52       ;2ND SIDE LENGTH ~
       Q220=+0        ;CORNER RADIUS ~
       Q368=+0        ;ALLOWANCE FOR SIDE ~
       Q224=+0        ;ANGLE OF ROTATION ~
       Q367=+0        ;POCKET POSITION ~
       Q207=+500      ;FEED RATE FOR MILLNG ~
       Q351=+1        ;CLIMB OR UP-CUT ~
       Q201=-1        ;DEPTH ~
       Q202=+1        ;PLUNGING DEPTH ~
       Q369=+0        ;ALLOWANCE FOR FLOOR ~
       Q206=+150      ;FEED RATE FOR PLNGNG ~
       Q338=+0        ;INFEED FOR FINISHING ~
       Q200=+5        ;SET-UP CLEARANCE ~
       Q203=+1        ;SURFACE COORDINATE ~
       Q204=+50       ;2ND SET-UP CLEARANCE ~
       Q370=+1        ;TOOL PATH OVERLAP ~
       Q366=+1        ;PLUNGE ~
       Q385=+500      ;FINISHING FEED RATE
29 CYCL CALL POS    X+0    Y+0    Z+0 FMAX
30 CYCL DEF 256 RECTANGULAR STUD ~
       Q218=+48       ;FIRST SIDE LENGTH ~
       Q424=+52       ;WORKPC. BLANK SIDE 1 ~
       Q219=+48       ;2ND SIDE LENGTH ~
       Q425=+52       ;WORKPC. BLANK SIDE 2 ~
       Q220=+0        ;CORNER RADIUS ~
       Q368=+0        ;ALLOWANCE FOR SIDE ~
       Q224=+0        ;ANGLE OF ROTATION ~
       Q367=+0        ;STUD POSITION ~
       Q207=+500      ;FEED RATE FOR MILLNG ~
       Q351=+1        ;CLIMB OR UP-CUT ~
       Q201=-15       ;DEPTH ~
       Q202=+3        ;PLUNGING DEPTH ~
       Q206=+3000 ;FEED RATE FOR PLNGNG ~
       Q200=+5        ;SET-UP CLEARANCE ~
       Q203=+0        ;SURFACE COORDINATE ~
       Q204=+50       ;2ND SET-UP CLEARANCE ~
       Q370=+1        ;TOOL PATH OVERLAP
31 CYCL CALL POS    X+0    Y+0    Z+0 FMAX
32 L    Z+200 F2000
33 L    X+35    Y+35
34 L    Z+10 F500
35 L    Z+0
36 LBL 1
37 L IZ-1 F500
38 L    X+20 RL F2000
39 L    Y-20
40 RND R5
41 L    X-20
42 CHF 6
43 L    Y+20
44 RND R5
45 L    X+0
46 CR    X+15 R+17 DR+
47 L    X+35
48 L    Y+35 R0
49 CALL LBL 1 REP6
50 L    Z+200 F2000
51 CYCL DEF 252 CIRCULAR POCKET ~
       Q215=+0        ;MACHINING OPERATION ~
       Q223=+20       ;CIRCLE DIAMETER ~
       Q368=+0        ;ALLOWANCE FOR SIDE ~
       Q207=+500      ;FEED RATE FOR MILLNG ~
       Q351=+1        ;CLIMB OR UP-CUT ~
       Q201=-5        ;DEPTH ~
       Q202=+2        ;PLUNGING DEPTH ~
       Q369=+0        ;ALLOWANCE FOR FLOOR ~
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      Q206=+150   ;FEED RATE FOR PLNGNG ~
      Q338=+0     ;INFEED FOR FINISHING ~
      Q200=+5     ;SET-UP CLEARANCE ~
      Q203=+0     ;SURFACE COORDINATE ~
      Q204=+50    ;2ND SET-UP CLEARANCE ~
      Q370=+1     ;TOOL PATH OVERLAP ~
      Q366=+1     ;PLUNGE ~
      Q385=+500   ;FINISHING FEED RATE
52 CYCL CALL POS   X+0   Y+0   Z+0 FMAX
53 L   Z+200 R0 F2000
54 CALL LBL 2
55 TOOL CALL 3 Z S800
56 CALL LBL 2
57 L   Z+200 R0 F1000
58 CALL LBL 3
59 CYCL DEF 200 DRILLING ~
      Q200=+5     ;SET-UP CLEARANCE ~
      Q201=-12    ;DEPTH ~
      Q206=+150   ;FEED RATE FOR PLNGNG ~
      Q202=+2     ;PLUNGING DEPTH ~
      Q210=+0     ;DWELL TIME AT TOP ~
      Q203=+0     ;SURFACE COORDINATE ~
      Q204=+50    ;2ND SET-UP CLEARANCE ~
      Q211=+0     ;DWELL TIME AT DEPTH
60 CYCL CALL POS   X+12   Y-12   Z+0 FMAX
61 CYCL CALL POS   X-12   Y-12   Z+0 FMAX
62 CYCL CALL POS   X-12   Y+12   Z+0 FMAX
63 CYCL CALL POS   X+12   Y+12   Z+0 FMAX
64 L   Z+200 R0 F2000
65 CALL LBL 2
66 M30
67 END PGM 1 MM
```